INTERMEDIATE LOGICS AND IMAGINARIES: THE TRANSLATION INTO AN INTERMEDIATE LOGIC COMMUTES WITH THE PRETOPOS COMPLETION OF ALL FINITE PROPOSITIONAL THEORIES EXACTLY FOR THE EXTENSIONS OF GÖDEL–DUMMETT LOGIC

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Abstract. For an intermediate logic L let $\mathsf{H}_L\mathcal{C}$ be the syntactic category of a coherent theory, with coherent syntactic category $\mathcal{C}$, read in first-order logic over L, and let P be the pretopos completion, which adds the imaginaries. There is a full and faithful comparison $\mathsf{H}_L\mathsf{P}\mathcal{C} \to \mathsf{P}\mathsf{H}_L\mathcal{C}$. For intuitionistic logic it is an equivalence only if $\mathcal{C}$ is Boolean. We determine how this depends on L. If L does not prove the weak excluded middle $\neg p \vee \neg\neg p$, the comparison is an equivalence exactly for the Boolean $\mathcal{C}$. If L proves it but not the linearity axiom $(p \to q) \vee (q \to p)$, the comparison fails for the theory of pure equality and for the propositional theory of the four-element chain. If L proves linearity, the comparison is an equivalence for every finite propositional theory. Hence the translation commutes with the imaginaries of all finite propositional theories if and only if L contains the Gödel–Dummett logic, and the same holds for all propositional theories; for the first-order theory of pure equality it commutes only for classical logic. The proofs use Kripke models over the fork and over the diamond, which are frames of L in the first and second cases, and in the third case the dual of the free L-algebra on a finite distributive lattice, the poset of strict chains of prime filters of bounded length, over which every finite presheaf embeds in one pulled back from the lattice.

1. Introduction

Let $\mathcal{C}$ be a small coherent category, the syntactic category of a coherent theory. Its pretopos completion $\mathsf{P}\mathcal{C}$ adds finite coproducts of sorts and quotients by definable equivalence relations, the imaginaries. In [6] we studied the free Heyting category $\mathsf{H}\mathcal{C}$ on $\mathcal{C}$, the theory read in intuitionistic first-order logic, and proved that the canonical comparison $\mathsf{H}\mathsf{P}\mathcal{C} \to \mathsf{P}\mathsf{H}\mathcal{C}$ is full and faithful and is an equivalence if and only if $\mathcal{C}$ is Boolean [6, Thm. 1.1]. For classical logic, on the other hand, the corresponding comparison is governed by a Galois criterion and is often an equivalence [7]. The proof for

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intuitionistic logic used a Kripke model over the fork, the poset $0 < 1$, $0 < 2$, and it was asked what happens for an intermediate logic, for which the fork need not be a frame [6, §9]. This paper answers the question.

An *intermediate logic* L is a consistent set of propositional formulas containing the intuitionistic theorems and closed under modus ponens and substitution. We write $\mathsf{H}_L\mathcal{C}$ for the syntactic category of the internal theory of $\mathcal{C}$ in intuitionistic first-order logic with all substitution instances of the formulas of L as axioms (Section 2), and

$$c^L_{\mathcal{C}} \colon \mathsf{H}_L\mathsf{P}\mathcal{C} \longrightarrow \mathsf{P}\mathsf{H}_L\mathcal{C}$$

for the canonical comparison, which is full and faithful (Lemma 2.2). A *finite propositional theory* is a finite distributive lattice D, regarded as a coherent category: the lattice of the coherent formulas of a propositional theory, up to provable equivalence, when this lattice is finite. The *weak excluded middle* is $\neg p \vee \neg\neg p$ and the *linearity axiom* is $(p \to q) \vee (q \to p)$; the logics they axiomatise are **KC** and the Gödel–Dummett logic $\mathbf{LC} \supseteq \mathbf{KC}$.

Theorem 1.1. *Let L be an intermediate logic.*

1. *If L does not prove the weak excluded middle, then for every small coherent category $\mathcal{C}$ the comparison $c^L_{\mathcal{C}}$ is an equivalence if and only if $\mathcal{C}$ is Boolean.*
2. *If L proves the weak excluded middle and does not prove the linearity axiom, then $c^L_{\mathcal{C}}$ is not an equivalence for the theory of pure equality and for the four-element chain; but the objects which obstruct in case* (1)*, two copies of an object glued along the negation of a subobject, are in the image.*
3. *If L proves the linearity axiom, then c^L_D is an equivalence for every finite distributive lattice D.*

In particular c^L_D is an equivalence for all finite distributive lattices D if and only if $L \supseteq \mathbf{LC}$.

Part (3) and the last assertion hold for all bounded distributive lattices, that is, for all propositional coherent theories (Theorem 6.3).

For first-order theories the situation is different: for the theory of pure equality $c^L_{\mathcal{C}}$ is an equivalence only when L is classical logic (Theorem 7.2).

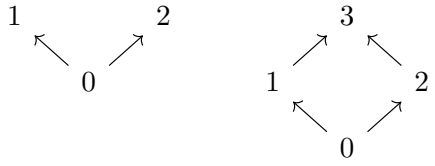


FIGURE 1. The fork and the diamond. An intermediate logic which does not contain **LC** has one of them as a frame (Lemma 3.1).

What is new and what is known. New, to the author's knowledge: Theorem 1.1, the embedding lemma for presheaves over strict chains (Lemma 6.2), and the obstruction over the diamond (Proposition 5.1), and the result for pure equality (Theorem 7.2). Known and used: the result for intuitionistic logic and its tools [6]; the duality between finite Heyting algebras and finite posets with p-morphisms [3]; local finiteness of the variety of Gödel algebras [5]; the pretopos completion as the category of coherent objects of the classifying topos [4, D3.3]. Lemma 3.1 is a case of Jankov's method of characteristic formulas [3]; we give the direct proof. The description of the free algebra in Proposition 6.1 is known: Gödel algebras free over finite distributive lattices are described by chains in [1], and free algebras over bounded distributive lattices in the subvarieties of Gödel algebras in [2]; we include a proof in the form needed.

2. The translation into an intermediate logic

Conventions are those of [6]: **Coh** is the $(2,1)$-category of small coherent categories, a Heyting category is a coherent category in which the pullback functors of subobjects have right adjoints $\forall_f$, a Heyting functor preserves them, $\mathrm{Mod}(\mathcal{C})$ is the category of models $\mathcal{C} \to \mathbf{Set}$ and homomorphisms, and universal properties are used also for targets such as $\mathbf{Set}^K$, the category of covariant functors from a poset K to sets.

An *L-algebra* is a Heyting algebra in which the formulas of L are valid. L-algebras are closed under subalgebras, quotients and products, and every Boolean algebra is an L-algebra, L being contained in classical logic. An *L-category* is a Heyting category $\mathcal{D}$ such that $\mathrm{Sub}(A)$ is an L-algebra for every object A. Boolean coherent categories are L-categories.

For $\mathcal{C}$ in **Coh** let $\mathsf{H}_L\mathcal{C}$ be the syntactic category of the internal coherent theory of $\mathcal{C}$ in intuitionistic first-order logic (with possibly empty sorts) with the additional axioms $\lambda(\varphi_1, \dots, \varphi_n)$, for $\lambda(p_1, \dots, p_n) \in L$ and first-order formulas φ_i in a common context. It is an L-category, its subobject lattices consisting of formulas in a context, and the unit $\eta\colon \mathcal{C} \to \mathsf{H}_L\mathcal{C}$ induces equivalences of groupoids between the Heyting functors $\mathsf{H}_L\mathcal{C} \to \mathcal{D}$ and the coherent functors $\mathcal{C} \to \mathcal{D}$ for every L-category $\mathcal{D}$: a coherent functor to a Heyting category interprets all first-order formulas soundly for intuitionistic logic, and the additional axioms are valid in an L-category. Every object of $\mathsf{H}_L\mathcal{C}$ is a subobject of some ηA. For intuitionistic logic, $\mathsf{H}_L = \mathsf{H}$.

Lemma 2.1. (a) *If $\mathcal{C}$ is Boolean, $\eta\colon \mathcal{C} \to \mathsf{H}_L\mathcal{C}$ is an equivalence.*

(b) *If $\mathcal{D}$ is an L-category, $\mathsf{P}\mathcal{D}$ is an L-category and a pretopos, the unit $\mathcal{D} \to \mathsf{P}\mathcal{D}$ is Heyting, and a coherent functor from $\mathsf{P}\mathcal{D}$ to a Heyting pretopos is Heyting iff its restriction to $\mathcal{D}$ is.*

Proof. (a) A coherent functor from a Boolean coherent category to a Heyting category is Heyting [6, Lemma 2.1], and $\mathcal{C}$ is an L-category; so the identity of $\mathcal{C}$ has the universal property of η. (b) Without the words "L-category" this is [6, Lemma 2.2]. An object of $\mathsf{P}\mathcal{D}$ is a quotient X/E of a finite coproduct

$X = \coprod_i A_i$ of objects of $\mathcal{D}$, and $\mathrm{Sub}(X/E)$ is the Heyting subalgebra of the E-saturated elements of $\mathrm{Sub}(X) \cong \prod_i \mathrm{Sub}(A_i)$; so it is an L-algebra. □

Lemma 2.2. *The Heyting functor $c^L_{\mathcal{C}} \colon \mathsf{H}_L\mathsf{P}\mathcal{C} \to \mathsf{P}\mathsf{H}_L\mathcal{C}$ extending $\mathsf{P}\eta$ is full and faithful, and it is an equivalence if and only if $\mathsf{H}_L\mathsf{P}\mathcal{C}$ is a pretopos.*

Proof. As [6, Lemma 2.3]: by Lemma 2.1(b) both $\mathsf{P}\mathsf{H}_L\mathsf{P}\mathcal{C}$ and $\mathsf{P}\mathsf{H}_L\mathcal{C}$ are the free L-category which is a pretopos on $\mathcal{C}$, and $c^L_{\mathcal{C}}$ is the unit $\mathsf{H}_L\mathsf{P}\mathcal{C} \to \mathsf{P}\mathsf{H}_L\mathsf{P}\mathcal{C}$ followed by the resulting equivalence. □

Kripke models over frames of L. A finite poset K is a *frame of L* if the Heyting algebra $\mathrm{Up}(K)$ of its up-sets is an L-algebra. An up-set $K' \subseteq K$ of a frame of L is a frame of L, restriction $\mathrm{Up}(K) \to \mathrm{Up}(K')$ being a surjective homomorphism.

Lemma 2.3. *Let K be a poset such that $\uparrow k$ is a finite frame of L for every $k \in K$. Then $\mathbf{Set}^K$ is an L-category. Hence every Kripke model $M \colon K \to \mathrm{Mod}(\mathcal{C})$ extends to Heyting functors M^{H_L}, $M^{\mathsf{H}_L\mathsf{P}}$, $M^{\mathsf{P}\mathsf{H}_L}$ to $\mathbf{Set}^K$ with $M^{\mathsf{P}\mathsf{H}_L}c^L_{\mathcal{C}} \cong M^{\mathsf{H}_L\mathsf{P}}$; and for every object X of $\mathsf{H}_L\mathsf{P}\mathcal{C}$ and all $k \leq l$ such that $M(k \leq l)$ is an isomorphism of models, the map $M^{\mathsf{H}_L\mathsf{P}}(X)(k) \to M^{\mathsf{H}_L\mathsf{P}}(X)(l)$ is injective.*

Proof. For a functor $A \colon K \to \mathbf{Set}$ the subobjects of A are the up-sets of the poset $\int A$ of pairs (k, x), $x \in A(k)$, with $(k, x) \leq (l, A(k \leq l)x)$; so $\mathrm{Sub}(A) = \mathrm{Up}(\int A)$. The restrictions $\mathrm{Up}(\int A) \to \mathrm{Up}(\uparrow(k, x))$ are homomorphisms of Heyting algebras, jointly injective, and $\uparrow(k, x) \cong \uparrow k$; so $\mathrm{Sub}(A)$ is a subalgebra of a product of L-algebras. The rest is as in [6, §3, Lemma 3.1]: X is a subobject of ηY with Y in $\mathsf{P}\mathcal{C}$, and the transition maps of $M^{\mathsf{P}}(Y)$ along isomorphisms of models are bijections. □

3. Which frames an intermediate logic has

Lemma 3.1. *Let L be an intermediate logic.*

(a) *The fork is a frame of L if and only if L does not prove the weak excluded middle.*
(b) *If L proves the weak excluded middle and not the linearity axiom, the diamond is a frame of L.*
(c) *If L proves the linearity axiom, a finite poset Z is a frame of L if and only if every $\uparrow z$ is a chain with at most n_L elements, where $n_L \in \{1, 2, \ldots, \infty\}$ is the supremum of the cardinalities of the finite chains which are frames of L.*

Proof. Let B be the Lindenbaum algebra of L, an L-algebra in which a formula is valid iff it is in L.

(a) In Up(fork) the weak excluded middle fails for $p = \{1\}$, since $\neg p = \{2\}$ and $\neg\neg p = \{1\}$. Conversely let $b \in B$ with $\neg b \vee \neg\neg b \neq 1$ and $c = \neg b$, so that $\neg\neg c = c$. The set $\{0, c, \neg c, c \vee \neg c, 1\}$ is a Heyting subalgebra of B: $c \wedge \neg c = 0$, $c \to \neg c = \neg c$, $\neg c \to c = \neg\neg c = c$, $(c \vee \neg c) \to c = c$, $(c \vee \neg c) \to \neg c = \neg c$

and $\neg(c \vee \neg c) = 0$. Its five elements are distinct because $c \vee \neg c \neq 1$, and it is isomorphic to Up(fork), with c and $\neg c$ the two atoms. So Up(fork) is an L-algebra.

(b) Let $a, b \in B$ with $(a \to b) \vee (b \to a) \neq 1$, and $x = a \to b$, $y = b \to a$, $t = x \wedge y$. In every Heyting algebra $x \to y = y$ and $y \to x = x$: if $z \wedge (a \to b) \leq b \to a$ then $z \wedge b \leq a$, as $b \leq a \to b$. Hence $\{t, x, y, x \vee y, 1\}$ is closed under $\wedge$, $\vee$ and $\to$: $(x \vee y) \to x = y \to x = x$, $(x \vee y) \to y = y$, $x \to t = x \to y = y$, $y \to t = x$ and $(x \vee y) \to t = x \wedge y = t$. Let $B' = B/F$ be the quotient by the filter F generated by $\neg\neg t$. In B' the class of $\neg t$ is 0, since $\neg t \wedge \neg\neg t = 0$; so the classes of $0, t, x, y, x \vee y, 1$ form a Heyting subalgebra G of B', the negation of each of $t, x, y, x \vee y$ being 0 in B'. We show that the class of $x \vee y$ is not 1, that is, $\neg\neg t \not\leq x \vee y$. In B the weak excluded middle gives $\neg t \vee \neg\neg t = 1$ and the De Morgan law $\neg(x \wedge y) = \neg x \vee \neg y$ (for $\neg(x \wedge y) \wedge \neg\neg x \leq \neg y$ in every Heyting algebra, and $\neg x \vee \neg\neg x = 1$). Now $\neg x = \neg(a \to b) = \neg\neg a \wedge \neg b \leq \neg b \leq y$ and likewise $\neg y \leq x$; so $\neg t \leq x \vee y$, and $\neg\neg t \leq x \vee y$ would give $x \vee y = 1$. It follows that the six classes are distinct: the classes of x and y are incomparable, since $[x] \leq [y]$ gives $[y] = [x \to y] = 1$; so $[t]$ is strictly below both and $[x \vee y]$ strictly above; and $[t] \neq 0$, since $[t] = 0$ iff $t \wedge \neg\neg t = 0$, while $t \wedge \neg\neg t = t$ and $t = 0$ would give $\neg\neg t = 0 \leq x \vee y$. The algebra G is isomorphic to Up(diamond), with $[t] = \{3\}$, $[x] = \{1, 3\}$, $[y] = \{2, 3\}$. So Up(diamond) is an L-algebra.

(c) Let Z be a finite frame of L and $z \in Z$. Then $\uparrow z$ is a frame of L; if it had incomparable elements a, b, the up-sets $p = \uparrow a$ and $q = \uparrow b$ would give $z \notin (p \to q) \cup (q \to p)$. So $\uparrow z$ is a chain, with at most n_L elements. Conversely a chain with $m \leq n_L$ elements is an up-set of a chain which is a frame of L, hence a frame of L; and $\mathrm{Up}(Z)$ is a subalgebra of the product of the $\mathrm{Up}(\uparrow z)$. □

4. Without weak excluded middle: only Boolean categories

Proof of Theorem 1.1(1). If $\mathcal{C}$ is Boolean, so is $\mathsf{P}\mathcal{C}$, and $\mathsf{H}_L\mathsf{P}\mathcal{C} \simeq \mathsf{P}\mathcal{C}$ is a pretopos by Lemma 2.1(a); so $c^L_{\mathcal{C}}$ is an equivalence by Lemma 2.2. Let $S \leq A$ be a subobject of $\mathcal{C}$ which is not complemented. By [6, Lemma 4.1] there are a homomorphism of models $h\colon M \to N$ and $a \in A(M) \setminus S(M)$ with $h_A(a) \in S(N)$. Let Q be the object of $\mathsf{PH}_L\mathcal{C}$ obtained from two copies of A by gluing along the pseudo-complement $\neg S$, and M^* the Kripke model over the fork with $M^*_0 = M^*_1 = M$, $M^*_2 = N$, the identity along $0 < 1$ and h along $0 < 2$. By Lemma 3.1(a) the fork and its up-sets are frames of L, so Lemma 2.3 applies. As in the proof of [6, Thm. 1.1], $a \in (\neg M^*(S))(1)$ and $a \notin (\neg M^*(S))(0)$, so the two copies of a are distinct points of $M^{*\mathsf{PH}_L}(Q)(0)$ with the same image at 1, which is impossible for an object in the image of $c^L_{\mathcal{C}}$. □

5. With weak excluded middle but without linearity

When L proves the weak excluded middle, the object Q of the preceding proof is in the image of $c^L_{\mathcal{C}}$: in $\mathrm{Sub}_{\mathsf{H}_L\mathcal{C}}(\eta A)$ one has $\neg S \vee \neg\neg S = A$ with $\neg S \wedge \neg\neg S = 0$, so A is the coproduct of $\neg S$ and $\neg\neg S$ in $\mathsf{PH}_L\mathcal{C}$, and

$$Q \;\cong\; \neg S + \neg\neg S + \neg\neg S,$$

a subobject of $A + A + A$ defined by formulas, hence the image of an object of $\mathsf{H}_L\mathsf{P}\mathcal{C}$. The obstruction moves from the negation to the implication.

Proposition 5.1. *Let L be an intermediate logic of which the diamond is a frame, and $\mathcal{C}$ a small coherent category. Suppose that there are subobjects $S, T \leq A$ in $\mathcal{C}$, homomorphisms of models $M \xrightarrow{h} N \xrightarrow{h'} N'$ and $a \in A(M)$ such that*

$$a \notin S(M), \qquad h(a) \in S(N) \setminus T(N), \qquad h'h(a) \in T(N').$$

Then $c^L_{\mathcal{C}}$ is not an equivalence: the object of $\mathsf{PH}_L\mathcal{C}$ obtained from two copies of A by gluing along $S \Rightarrow T$, the Heyting implication in $\mathrm{Sub}_{\mathsf{H}_L\mathcal{C}}(\eta A)$, is not in its essential image.

Proof. Let $U = (\eta S \Rightarrow \eta T) \leq \eta A$ in $\mathsf{H}_L\mathcal{C}$ and $Q = (A+A)/E$ in $\mathsf{PH}_L\mathcal{C}$, with E the equivalence relation on $A+A$ which is the union of the diagonal and of the two images of U in the off-diagonal summands $A \times A$ of $(A+A)\times(A+A)$, as in [6, §5]. Let M^* be the Kripke model over the diamond with

$$M^*_0 = M^*_1 = M, \qquad M^*_2 = N, \qquad M^*_3 = N',$$

the identity along $0 < 1$, h along $0 < 2$, h' along $2 < 3$ and $h'h$ along $1 < 3$. The up-sets of the diamond are frames of L, so Lemma 2.3 applies, and $M^{*\mathsf{PH}_L}(Q)(k)$ is the quotient of two copies of $A(M^*_k)$ identifying the copies of the points of $U_{M^*}(k)$, where, the implication of $\mathbf{Set}^K$ being the Kripke implication,

$$U_{M^*}(k) = \{x \mid \text{for all } l \geq k,\ x|_l \in S(M^*_l) \text{ implies } x|_l \in T(M^*_l)\}.$$

For $k = 1$ and $x = a$ the nodes $l \geq 1$ are 1, where $a \notin S(M)$, and 3, where $h'h(a) \in T(N')$; so $a \in U_{M^*}(1)$. For $k = 0$ the node $l = 2$ has $h(a) \in S(N) \setminus T(N)$; so $a \notin U_{M^*}(0)$. Hence the two copies of a are distinct at 0 and equal at 1, along a transition which is an isomorphism of models, and Q is not in the essential image of $c^L_{\mathcal{C}}$ by Lemma 2.3. □

Proof of Theorem 1.1(2). By Lemma 3.1(b) the diamond is a frame of L. For the theory of pure equality, with one sort σ, take $A = \sigma^3$, $S = [x_1 = x_2]$, $T = [x_1 = x_2 \wedge x_2 = x_3]$, a three-element set M with a an enumeration of it, and maps $M \to N \to N'$ onto a two-element set, identifying a_1 and a_2, and then onto a one-element set. For the four-element chain $0 < q < p < 1$, a propositional theory, take $A = 1$, $S = p$, $T = q$ and the prime filters $\{1\} \subset \{p, 1\} \subset \{q, p, 1\}$. Proposition 5.1 applies in both cases. The statement on the objects of case (1) was shown before the proposition. □

6. With linearity: finite propositional theories

Throughout this section L proves the linearity axiom and $n = n_L$ is as in Lemma 3.1(c).

Finite duality. For a finite poset X, $\mathrm{Up}(X)$ is a finite Heyting algebra, and every finite Heyting algebra is of this form, with X the poset of its prime filters. A monotone map $f\colon Z \to X$ induces the lattice homomorphism $f^{-1}\colon \mathrm{Up}(X) \to \mathrm{Up}(Z)$, and every homomorphism of bounded lattices $\mathrm{Up}(X) \to \mathrm{Up}(Z)$ is of this form for a unique f; f^{-1} is a homomorphism of Heyting algebras if and only if f is a *p-morphism*, $f(\uparrow z) = \uparrow f(z)$ for all z [3]. A finite distributive lattice D is $\mathrm{Up}(X_D)$ for the poset X_D of its prime filters, which are the models of D as a coherent category, ordered by inclusion, the homomorphisms.

For a finite poset X let $\mathrm{Ch}_n(X)$ be the set of the nonempty strict chains $c = (c_0 < c_1 < \cdots < c_m)$ of X with at most n elements, ordered by $c \le c'$ iff c' is a final segment of c, and $\pi\colon \mathrm{Ch}_n(X) \to X$, $\pi(c) = c_0$, a monotone map. The up-set of c is the chain of its final segments.

Proposition 6.1. *For a finite distributive lattice $D = \mathrm{Up}(X)$, the free L-algebra on the bounded lattice D is $\mathrm{Up}(\mathrm{Ch}_n(X))$ with $\pi^{-1}\colon D \to \mathrm{Up}(\mathrm{Ch}_n(X))$; and $\mathsf{H}_L D$ is equivalent to it, as a preorder.*

Proof. $\mathrm{Up}(\mathrm{Ch}_n(X))$ is an L-algebra by Lemma 3.1(c). Let $\mathrm{Up}(Z)$ be a finite L-algebra and $f\colon Z \to X$ monotone. For $z \in Z$ the up-set $\uparrow z$ is a chain $z = z_0 < \cdots < z_r$ with at most n elements, and $f(z_0) \le \cdots \le f(z_r)$; let $g(z)$ be the strict chain of the distinct values. Then $g\colon Z \to \mathrm{Ch}_n(X)$ is a p-morphism with $\pi g = f$: the final segments of $g(z)$ are the $g(z_i)$. It is the only one: if g' is a p-morphism with $\pi g' = f$, the elements of the chain $g'(z)$ are the $\pi(c')$, $c' \ge g'(z)$, that is the $\pi g'(z') = f(z')$, $z' \ge z$. So $\mathrm{Up}(\mathrm{Ch}_n(X))$ has the universal property for finite L-algebras; applied to the subalgebra generated by $\pi^{-1}(D)$ and to itself, it shows that $\pi^{-1}(D)$ generates it. An arbitrary L-algebra is a Gödel algebra, and Gödel algebras are locally finite [5]; so a lattice homomorphism from D to an L-algebra has its image in a finite subalgebra, and the universal property holds for all L-algebras. Finally every object of $\mathsf{H}_L D$ is a subobject of some ηd, a subterminal object; so $\mathsf{H}_L D$ is a preorder, its poset of subterminal objects is an L-algebra, an L-algebra is an L-category, and Heyting functors between Heyting algebras are homomorphisms; the universal properties agree. □

Pretopos completions. For a finite poset X the category $\mathbf{FinSet}^X$ of functors $X \to \mathbf{FinSet}$ is a pretopos, the functor $\mathrm{Up}(X) \to \mathbf{FinSet}^X$ sending an up-set to its characteristic subterminal functor is coherent, and it is the pretopos completion of $\mathrm{Up}(X)$: $\mathbf{Set}^X$ is the topos of sheaves on X with the topology of up-sets, the classifying topos of the propositional theory $\mathrm{Up}(X)$, and its coherent objects are the finite functors [4, D3.3]. For $f\colon Z \to X$ the functor $\mathsf{P}(f^{-1})$ is $f^*\colon \mathbf{FinSet}^X \to \mathbf{FinSet}^Z$, composition with f. Every Y

in $\mathbf{FinSet}^X$ is covered by subterminal objects: $y \in Y(x)$ defines a morphism $s_y\colon \uparrow x \to Y$, and the s_y are jointly surjective.

So, with $Z = \mathrm{Ch}_n(X)$,

$$\mathsf{P}D = \mathbf{FinSet}^X, \qquad \mathsf{PH}_L D = \mathbf{FinSet}^Z, \qquad \mathsf{P}\eta = \pi^*,$$

and $\mathbf{FinSet}^Z$ is an L-category by Lemma 2.3.

Lemma 6.2 (embedding)**.** *Every W in $\mathbf{FinSet}^Z$, $Z = \mathrm{Ch}_n(X)$, is a subobject of π^*Y for some Y in $\mathbf{FinSet}^X$; namely for the left Kan extension Y of W along π, the unit $W \to \pi^*Y$ is a monomorphism.*

Proof. $Y(x)$ is the colimit of W over the poset $(\pi \downarrow x)$ of the chains c with $c_0 \le x$. For such c the elements $c_i \le x$ form an initial segment of c; let $\tau_x(c)$ be the final segment of c beginning at the last of them. The chains $c' \ge c$ in $(\pi \downarrow x)$ are the final segments of c beginning at some $c_i \le x$; so $\tau_x(c)$ is the greatest of them and $\tau_x(c') = \tau_x(c)$. Hence each connected component of $(\pi \downarrow x)$ has a greatest element, and

$$Y(x) = \coprod_{r \in R_x} W(r), \qquad R_x = \{c \mid c_0 \le x, \text{ and } c = (c_0) \text{ or } c_1 \not\le x\},$$

a finite set, the map $W(c) \to Y(x)$ being $W(c \le \tau_x(c))$ into the summand of $\tau_x(c)$. For $x = \pi(c) = c_0$ the chain c is in R_x, because $c = (c_0)$ or $c_1 > c_0$; so the unit $W(c) \to Y(\pi(c))$ is a coproduct injection. □

Lemma 6.2 is a combinatorial fact about the posets of chains. Linearity is used in Proposition 6.1, which identifies the dual of the free L-algebra with such a poset; for a logic which does not prove linearity the up-sets of the points of a finite frame need not be chains, and greatest elements of this kind are not available.

Proof of Theorem 1.1(3). By Lemma 2.2 it suffices to show that $c = c^L_D\colon \mathsf{H}_L\mathsf{P}D \to \mathsf{PH}_L D = \mathbf{FinSet}^Z$ is essentially surjective. We have $c\,\eta_{\mathsf{P}D} \cong \pi^*$, and $c \circ \mathsf{H}_L(\iota) \cong \iota_{\mathsf{H}_L D}$, where ι denotes the units of P, both sides being Heyting functors $\mathsf{H}_L D \to \mathbf{FinSet}^Z$ extending the same coherent functor on D.

Let W be in $\mathbf{FinSet}^Z$. By Lemma 6.2, W is a subobject of $\pi^*Y = c(\eta Y)$ for some Y in $\mathbf{FinSet}^X = \mathsf{P}D$. Let $s_j\colon u_j \to Y$, $j = 1, \dots, m$, be morphisms of $\mathsf{P}D$ from subterminal objects $u_j \in D$ which are jointly surjective. For each j the pullback $\varphi_j := (\pi^* s_j)^* W$ is a subobject of $\pi^* u_j$ in $\mathbf{FinSet}^Z$, a subterminal object, that is an element of $\mathrm{Up}(Z) = \mathsf{H}_L D$ below $\pi^{-1}(u_j)$; let $\varphi'_j \le \eta u_j$ be its image under $\mathsf{H}_L(\iota)\colon \mathsf{H}_L D \to \mathsf{H}_L\mathsf{P}D$, and

$$X := \bigvee_j \exists_{\eta s_j} \varphi'_j \;\le\; \eta Y$$

in $\mathsf{H}_L\mathsf{P}D$. The functor c preserves images and unions and sends φ'_j to φ_j, so $c(X) = \bigvee_j \exists_{\pi^* s_j} (\pi^* s_j)^* W = W$, the $\pi^* s_j$ being jointly surjective. □

Proof of the last assertion of Theorem 1.1. If $L \supseteq \mathbf{LC}$, part (3) applies. If not, L does not prove linearity; by parts (1) and (2), c^L_D is not an equivalence for the four-element chain D, which is not Boolean. □

Arbitrary propositional theories. The finiteness of D can be removed, because every object of the pretopos completion of a lattice is given by finitely many elements.

Theorem 6.3. *If L proves the linearity axiom, then c^L_D is an equivalence for every bounded distributive lattice D. Hence the translation into L commutes with the pretopos completion of all propositional coherent theories if and only if $L \supseteq \mathbf{LC}$.*

Proof. By Lemma 2.2 it suffices to show that c^L_D is essentially surjective. As in Proposition 6.1, $\mathsf{H}_L D$ is, as a preorder, the free L-algebra G on the bounded lattice D. For a finite sublattice $D_i \subseteq D$ containing 0 and 1, with inclusion j, let G_i be the free L-algebra on D_i and $j_* \colon G_i \to G$ the induced homomorphism. The lattice D is the directed union of its finite sublattices, a directed colimit of bounded lattices; the free functor preserves colimits, and directed colimits of algebras are computed on the underlying sets. So every element of G is $j_*(v)$ for some finite D_i and $v \in G_i$, and if finitely many inequalities between elements $j_*(v)$, $v \in G_i$, hold in G, the corresponding inequalities hold in $G_{i'}$ for some finite $D_{i'} \supseteq D_i$.

An object W of $\mathsf{P}G$ is the quotient of a finite coproduct $\coprod_k u_k$ of subterminal objects $u_k \in G$ by an equivalence relation, a family of elements $e_{kl} \leq u_k \wedge u_l$ of G with $e_{kk} = u_k$, $e_{kl} = e_{lk}$ and $e_{kl} \wedge e_{lm} \leq e_{km}$. By the preceding paragraph there are a finite sublattice D_i and elements of G_i with the same properties which are sent to the u_k and the e_{kl}; they define an object W_i of $\mathsf{P}G_i$ with $\mathsf{P}(j_*)(W_i) \cong W$, the functor $\mathsf{P}(j_*)$ being coherent between pretoposes. By Theorem 1.1(3), $W_i \cong c^L_{D_i}(X_i)$ for an object X_i of $\mathsf{H}_L\mathsf{P}D_i$. Finally

$$c^L_D \circ \mathsf{H}_L\mathsf{P}(j) \;\cong\; \mathsf{P}\mathsf{H}_L(j) \circ c^L_{D_i},$$

both sides being Heyting functors $\mathsf{H}_L\mathsf{P}D_i \to \mathsf{P}\mathsf{H}_L D$ whose restrictions to $\mathsf{P}D_i$ are coherent functors into a pretopos restricting to ηj on D_i; and $\mathsf{P}\mathsf{H}_L(j) = \mathsf{P}(j_*)$. So $W \cong c^L_D(\mathsf{H}_L\mathsf{P}(j)X_i)$. The second assertion follows from the last assertion of Theorem 1.1. □

7. First-order theories: pure equality only for classical logic

The positive result of the preceding section does not extend to first-order theories. Already for the theory without symbols and axioms the translation commutes with the imaginaries only when the logic is classical. The obstruction is the quotient of the sort by the double negation of equality, and the frame is the two-element chain.

Lemma 7.1. *The two-element chain $0 < 1$ is a frame of every intermediate logic other than classical logic.*

Proof. If L is not classical, its Lindenbaum algebra has an element b with $d := b \vee \neg b \neq 1$. Then $\neg d = \neg b \wedge \neg\neg b = 0$, so $\{0, d, 1\}$ is a Heyting subalgebra, isomorphic to the algebra of the up-sets of the two-element chain. □

Theorem 7.2. *Let $\mathcal{C}$ be the syntactic category of the coherent theory of pure equality, with one sort σ and no symbols and axioms. For an intermediate logic L the comparison $c_{\mathcal{C}}^{L}$ is an equivalence if and only if L is classical logic.*

Proof. For classical logic H_L is the Morleyization and the statement is [7, Ex. 6.5]. Let L be non-classical, and K the chain $0 < 1$, a frame of L with its up-sets (Lemma 7.1). The models of $\mathcal{C}$ are the sets and the homomorphisms are all maps. Let Ω be a countably infinite set with a partition into at least two blocks, all of them infinite, $p\colon \Omega \to \Omega/\!\sim$ the projection to the set of blocks, and M the Kripke model over K with $M_0 = \Omega$, $M_1 = \Omega/\!\sim$ and transition p.

In every Heyting category $\neg\neg(x = y)$ is an equivalence relation on σ; let $W = \sigma/\neg\neg(x = y)$ in $\mathsf{PH}_L\mathcal{C}$. In $\mathbf{Set}^K$ a pair (a, b) of elements of Ω satisfies $\neg\neg(x = y)$ at 0 iff $p(a) = p(b)$, by the formula for the Kripke negation; so $M^{\mathsf{PH}_L}(W)(0)$ is the set of blocks.

Suppose that $W \cong c_{\mathcal{C}}^{L}(X)$ with $X \leq \eta Y$ in $\mathsf{H}_L\mathsf{P}\mathcal{C}$, Y in $\mathsf{P}\mathcal{C}$. Write $Y = (\coprod_i A_i)/F$ with objects A_i of $\mathcal{C}$, definable subsets of powers σ^{n_i} in the language of equality, and F a coherent equivalence relation. By the compatibility of the extensions of M with $c_{\mathcal{C}}^{L}$ (Lemma 2.3) and since $X \leq \eta Y$, we obtain an injection

$$\begin{aligned}\theta\colon \{\text{blocks}\} = M^{\mathsf{PH}_L}(W)(0) \;\cong\; M^{\mathsf{H}_L\mathsf{P}}(X)(0) \\ \subseteq\; Y(\Omega) = \Big(\coprod_i A_i(\Omega)\Big)/F(\Omega).\end{aligned}$$

Let G be the group of the permutations g of Ω which preserve the partition, in the sense that $a \sim b$ iff $g(a) \sim g(b)$; they may permute the blocks. Each $g \in G$, with the induced permutation of the blocks, is an automorphism of the Kripke model M; it induces natural automorphisms of the extensions of M (an extension together with its identification with M on $\mathcal{C}$ being unique up to a unique isomorphism), and the isomorphism $W \cong c_{\mathcal{C}}^{L}(X)$ is natural with respect to them. So θ is G-equivariant, G acting on the blocks and on $Y(\Omega)$ in the evident way. Moreover every permutation π of Ω is an automorphism of the model Ω and preserves $F(\Omega)$.

Let B be a block and $\theta(B)$ the class of a tuple $d \in A_i(\Omega)$. Since the blocks are infinite there is $g \in G$ mapping every block to itself with $g(d_j) \notin \{d_1, \ldots, d_{n_i}\}$ for all j; then $gB = B$, so $F(d, gd)$. Let $B' \neq B$ be another block; there is $\varepsilon \in G$ exchanging B and B', mapping the other blocks to themselves, with $\varepsilon(d_j) \notin \{d_1, \ldots, d_{n_i}\}$ for all j. The map $d_j \mapsto d_j$, $g(d_j) \mapsto \varepsilon(d_j)$ is a well defined injective map between finite subsets of Ω, so it extends to a permutation π of Ω, and $F(d, gd)$ gives $F(\pi d, \pi g d) = F(d, \varepsilon d)$. Hence $\varepsilon\theta(B) = \theta(B)$, that is $\theta(\varepsilon B) = \theta(B)$ with $\varepsilon B = B' \neq B$, contradicting

the injectivity of θ. (If $n_i = 0$ the class $\theta(B)$ is fixed by G, with the same contradiction.) □

The transition p of this Kripke model is not an isomorphism, and the invariant of Lemma 2.3 is not what excludes W; the argument is the one which shows in [7, Thm. 5.3] that a classical imaginary is not coherent, an automorphism which moves a class and cannot be told from one which fixes it by a coherent formula. For propositional theories there are no such automorphisms, which is consistent with Theorem 1.1(3).

8. Summary and questions

assumption on L	frame used	$c^L_{\mathcal{C}}$ is an equivalence
no w.e.m.	fork	iff $\mathcal{C}$ is Boolean
w.e.m., no linearity	diamond	not for pure equality, the 4-chain
linearity	chains ($\leq n_L$)	for all bounded distributive lattices
not classical	two-element chain	not for pure equality
classical logic	point	iff the criterion of [7, Thm. 6.1]

Here w.e.m. is the weak excluded middle. The last line is included for comparison: for classical logic H_L is the Morleyization, every finite propositional theory satisfies the criterion [7, Prop. 5.2], and the theory of pure equality satisfies it as well [7, Ex. 6.5].

Question 8.1. Let $L \supseteq \mathbf{LC}$ be different from classical logic. By Theorem 7.2 the comparison fails for pure equality. For which coherent categories $\mathcal{C}$ is $c^L_{\mathcal{C}}$ an equivalence? For the three-element chain D the pretopos $\mathcal{C} = \mathsf{P}D$ is an example which is neither Boolean nor a preorder, by Theorem 1.1(3); but it is Morita equivalent to the propositional theory D. Are there examples which are neither Boolean nor Morita equivalent to a propositional theory, that is, with $\mathsf{P}\mathcal{C}$ not equivalent to $\mathsf{P}D$ for any distributive lattice D?

Question 8.2. In case (2), for which non-Boolean $\mathcal{C}$ is $c^L_{\mathcal{C}}$ an equivalence? The three-element chain has no configuration as in Proposition 5.1.

Declarations

The author has no competing interests to declare.

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